

SYNOPSIS

YEAR/BRANCH.: BE(E&TC)

DIVISION: A

GROUP NO.: A26

TITLE: Smart Industrial Monitoring System with Predictive Maintenance Using CAN FD Network and Edge Machine Learning Analytics.

OBJECTIVES:

- To design and implement a high-speed CAN FD network for industrial sensor data communication
- To develop edge computing-based machine learning models for real-time anomaly detection and predictive maintenance
- To integrate temperature, vibration, current, and power sensors for comprehensive equipment health monitoring
- To create a cloud-connected real-time dashboard with alerting and historical data visualization capabilities

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INTRODUCTION:

The rise of Industry 4.0 accentuates the need for intelligent, reliable, and real-time monitoring systems that reduce downtime and optimize maintenance costs. Traditional maintenance strategies—reactive and preventive—are costly and inefficient due to unplanned failures or premature servicing. This project aims to address these challenges by developing a smart industrial monitoring system that leverages the latest advancements in CAN FD networking and edge machine learning analytics. The system collects multi-parameter sensor data from industrial

equipment through ESP32 microcontrollers connected over a CAN FD bus to a Raspberry Pi-based gateway. The gateway processes this data using machine learning algorithms to detect anomalies and predict failures, thus enabling predictive maintenance. A cloud-based dashboard provides remote access, real-time alerts, and historical insights to facilitate operational decision-making. This approach aims to improve equipment uptime, reduce maintenance cost, and enhance overall production efficiency.

BLOCK DIAGRAM:

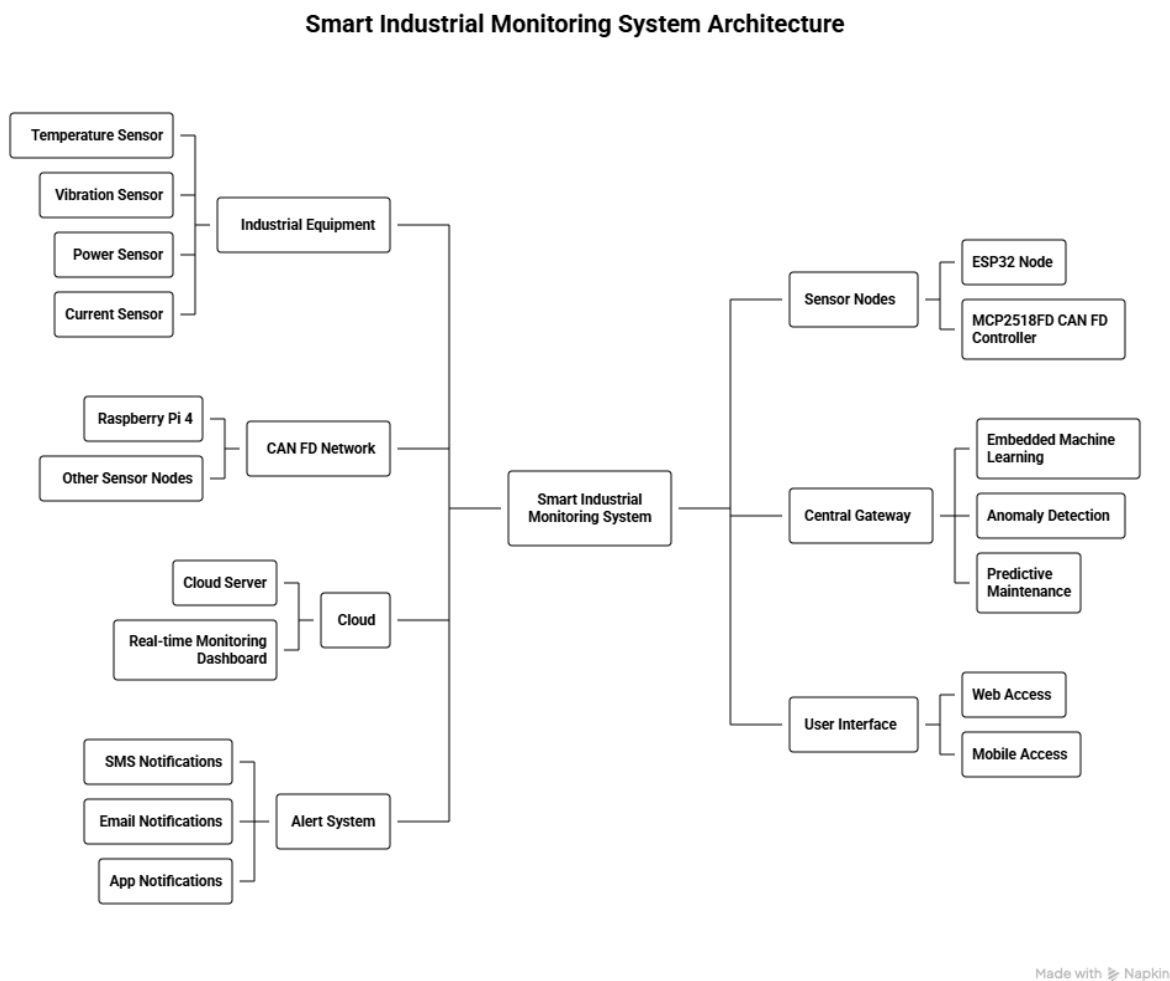


Figure 1: Block Diagram of Smart Industrial Monitor System

BLOCK DIAGRAM DESCRIPTION:

The block diagram illustrates the overall architecture of the system. Multiple sensor nodes consisting of ESP32 microcontrollers interfaced with temperature, vibration, and current sensors

collect real-time data from industrial machines. These nodes communicate via the CAN FD network, providing high-speed, reliable data transmission. The Raspberry Pi gateway acts as an edge computing device, running machine learning algorithms to analyze sensor data locally, reducing latency for anomaly detection and predictive maintenance notifications. Processed data and alerts are simultaneously sent to a cloud-based dashboard accessible remotely by engineers and operators, ensuring timely maintenance actions.

ADVANTAGES & APPLICATIONS:

- Enables real-time and predictive maintenance, reducing unplanned machine downtime.
- Improves operational efficiency by preventing catastrophic equipment failures.
- Supports integration with existing industrial systems via CAN FD protocol.
- Cloud connectivity allows remote monitoring, increasing accessibility and response time.
- Scalable system architecture suitable for various industrial automation scenarios.
- Applicable in manufacturing, power generation, automotive assembly, and process industries.

CONCLUSION: This project demonstrates the design and implementation of a smart industrial monitoring system that integrates advanced CAN FD communication and edge machine learning for predictive maintenance. The system is expected to significantly reduce maintenance costs and increase equipment availability through timely detection of faults, showcasing the potential of Industry 4.0 technologies for modern manufacturing.

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